

Multiple Choice

1. **Answer:** C

Options: A) 190; B) 200; C) 170; D) 180; E) 120 *Note: Correct option: C - 170*

1. **Answer:** C

Options: A) 0; B) 1; C) 2; D) 10

Note: Correct option: C - 2

2. **Answer:** A

Options: A) 0.015 joule.; B) 15 joule.; C) 0.0015 joule.; D) 1.5 joule.; E) 0.15 millijoule.

Note: Correct option: A - 0.015 joule.

3. **Answer:** A

Options: A) 2.00 bits/message; B) 2.25 bits/message; C) 1.75 bits/message; D) 3.00 bits/message; E) 2.75 bits/message

Note: Correct option: A - 2.00 bits/message

4. **Answer:** A

Options: A) $f_1(t) \{_{st}\} f_2(t) = (1.6 \cos 3t + 1.2 \sin 3t - 1.6e^{-4t})u(t)$; B) $f_1(t) \{_{st}\} f_2(t) = (1.2 \cos 3t + 1.6 \sin 3t - 1.3e^{-4t})u(t)$; C) $f_1(t) \{_{st}\} f_2(t) = (1.4 \cos 3t + 1.2 \sin 3t - 1.5e^{-4t})u(t)$; D) $f_1(t) \{_{st}\} f_2(t) = (1.6 \cos 3t + 1.0 \sin 3t - 1.6e^{-4t})u(t)$

Note: Correct option : A - $f_1(t) \{_{st}\} f_2(t) = (1.6 \cos 3t + 1.2 \sin 3t - 1.6e^{-4t})u(t)$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $3z^2 + 4z^{-1} + 2z^{-3}$ '.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $F(s) = [(1/2) / (s+1)^2] + [(1/4) / (s+1)] - [(1/4) / (s+3)]$ '.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Process box.'.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $(1/2) \text{ erg/meter}^3$ '.

Long Form Response

11. **Answer:** 0.7 meters, 1 meter. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 2. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key